Inference Investigation (HetGP)

Harry Newton

2026-06-11

1 Test Problem

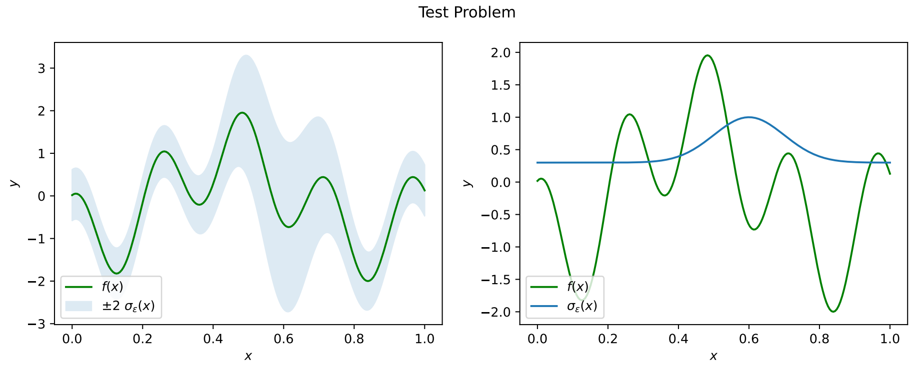
The VHGP is fitted to a heteroscedastic test function shown in Figure 1. The dataset is composed of $k = 10$ unique locations, with $n = 5$ replicates taken at each point.

Table 1: Summary of experiment's parameter values.

Parameter	Value
k	10
n	5
seed	12345

```
Text(0.5, 0.98, 'Test Problem')
```

(a) The target function of which noisy data we condition our HetGP on.



(b)

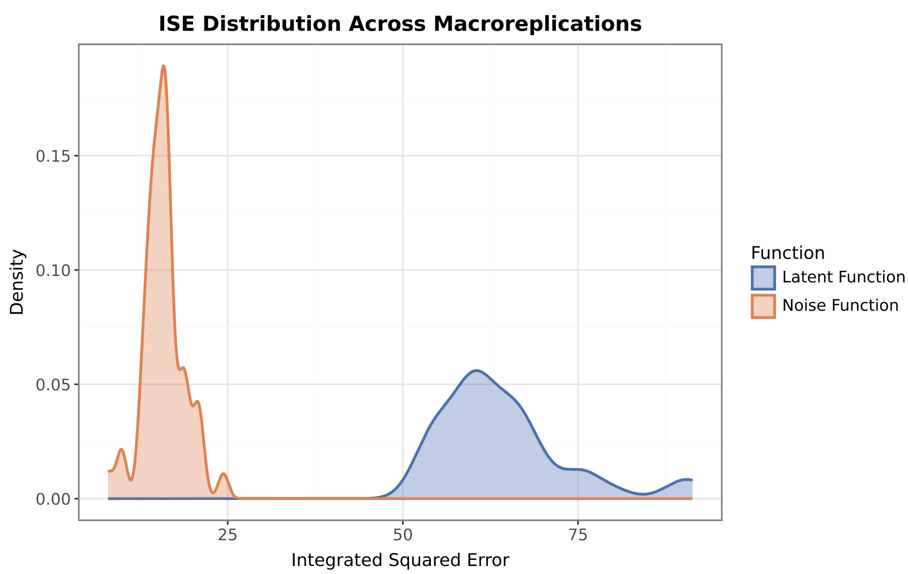
Figure 1

2 Results

Included in this section is the table Table 2; summarising the model likelihood, the ideal likelihood - the likelihood if the truth was our model -, and the integrated square errors for both the noise and latent predictions.

Table 2: Summary of log-likelihoods and integrated square errors

	ll	ideal_ll	ISE_f	ISE_eps
count	50.000000	50.000000	50.000000	50.000000
mean	-28.811505	-26.289748	64.368391	15.802749
std	8.043122	5.314669	9.284083	2.957538
min	-45.870932	-39.080010	52.625051	7.920439
25%	-35.289392	-28.814394	58.953868	14.337610
50%	-27.877813	-25.827286	61.938543	15.857148
75%	-23.294950	-23.349741	67.243647	16.734347
max	-12.281763	-15.028923	91.342903	24.390810

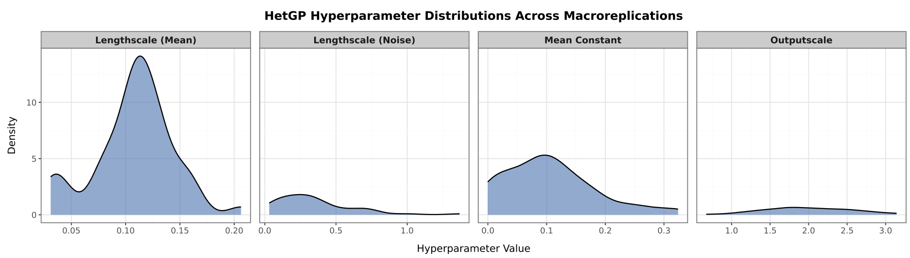


(a)

Figure 2: The Integrated Square Error (ISE) for both noise and latent function, after $M = 50$ macroreplications.

Table 3: Summary stats for HetGP hyperparameters

	l_f	l_g	tau	mu
count	50.000000	50.000000	50.000000	50.000000
mean	0.107674	0.344038	2.004278	0.105134
std	0.036719	0.269688	0.555587	0.078212
min	0.030971	0.030971	0.671365	-0.000253
25%	0.090723	0.153430	1.670276	0.047919
50%	0.112074	0.289896	1.882544	0.097785
75%	0.128326	0.445295	2.438337	0.145029
max	0.206283	1.368036	3.142460	0.324011



(a)

Figure 3: A plot of hyperparameter densities after 50 macroreplications.

3 Appendix

Table 4: Contains the raw hyperparameter data from the experiments

l_f	l_g	tau	mu
0.103817	0.296325	1.397666	0.068805
0.112833	0.454598	1.824916	0.095564
0.111157	0.205902	1.762399	0.084664
0.068903	0.294711	2.054674	0.024018
0.123244	0.386164	1.439141	0.145322
0.084276	0.091856	2.625707	0.024933
0.104821	0.731323	1.867648	0.075225
0.127451	0.276926	2.763088	0.172756
0.133589	1.368036	1.267703	0.127848
0.090017	0.343780	1.831740	0.049423
0.099811	0.550780	1.773285	0.047418
0.115813	0.660447	2.423278	0.102968
0.112769	0.193467	1.560573	0.107016
0.083627	0.167986	1.783092	0.026615
0.098317	0.106792	2.120604	0.073785
0.127059	0.339004	2.717993	0.144150
0.046550	0.046550	2.443357	0.003958
0.111429	0.741910	2.505663	0.103005
0.113981	0.149515	2.130161	0.107634
0.129247	0.300205	2.455855	0.179966
0.105109	0.364913	1.679801	0.084484
0.030971	0.030971	3.107290	-0.000253
0.206283	0.244068	2.170641	0.324011
0.043885	0.043948	0.671365	0.002450
0.097449	0.147172	1.794746	0.074978
0.165816	0.534731	2.300202	0.256791
0.104622	0.756577	1.473972	0.077390
0.120383	0.271831	3.142460	0.124373
0.115124	0.146729	2.596065	0.116349
0.131004	0.210723	2.294375	0.145393
0.128618	0.191982	1.643014	0.130401
0.092840	0.101878	1.807906	0.057450
0.031302	0.031437	1.790004	-0.000210
0.158918	0.369716	3.051749	0.302446
0.122631	0.569414	1.694163	0.118524
0.080982	0.273902	1.667101	0.023901
0.144480	1.001256	1.897440	0.164042
0.030971	0.030971	2.643863	0.000563
0.151414	0.229257	2.117266	0.186291
0.153797	0.165177	2.488221	0.236599
0.076221	0.112796	1.303563	0.036057
0.136800	0.446952	2.123604	0.177798
0.136930	0.356796	2.295301	0.180707
0.115896	0.377371	2.262179	0.100006
0.159141	0.285081	2.728414	0.240028
0.035677	0.043342	1.234713	0.001371
0.110101	0.343431	1.178804	0.088222
0.076209	0.673671	1.331443	0.027223
0.112720	0.440327	1.827898	0.122094
0.108680	0.699211	1.147811	0.092145

Table 5: Contains the raw metric data from the experiments

m	ll	ideal_ll	ISE_f	ISE_eps
0	-35.402183	-23.482327	63.251027	14.513199
1	-34.202205	-25.092985	60.397434	14.647721
2	-29.577056	-24.109592	57.859055	15.893857
3	-15.463001	-15.028923	68.442447	20.849712
4	-27.299614	-23.552763	54.536925	13.671453
5	-13.484628	-25.017320	64.419766	18.814434
6	-38.292985	-28.865564	59.149202	15.064309
7	-20.075230	-19.560612	60.239300	16.777097
8	-42.099616	-26.825102	67.416689	24.390810
9	-27.417352	-25.747501	60.812753	15.925255
10	-28.537324	-25.299360	64.105671	15.724515
11	-36.592346	-25.730337	55.532840	14.686465
12	-36.565669	-31.712303	52.625051	12.549599
13	-29.874435	-25.714716	59.910076	16.029084
14	-27.737320	-25.949308	65.817785	15.820438
15	-22.319736	-20.650100	56.063794	15.142150
16	-21.639647	-28.660883	66.088996	18.736305
17	-30.638345	-21.744133	58.906381	14.594117
18	-12.281763	-15.143691	56.271464	20.827700
19	-20.517291	-30.255838	54.214928	18.166734
20	-35.610593	-24.208761	73.587328	14.511918
21	-23.230306	-29.259533	88.563708	19.169226
22	-38.846424	-36.800570	75.264458	10.024767
23	-15.233859	-22.123613	76.869029	20.325700
24	-26.982489	-27.262516	66.343377	16.464841
25	-36.809068	-31.116812	60.097377	13.094840
26	-22.026283	-21.105830	57.538581	9.671193
27	-16.781620	-20.596405	69.377477	17.803069
28	-24.711307	-24.586615	55.476304	16.021607
29	-27.348919	-25.907072	61.660590	13.781946
30	-30.725336	-27.922226	66.697174	14.164394
31	-24.515543	-20.130112	52.816464	17.132591
32	-23.488885	-26.863308	91.186545	19.048555
33	-26.863419	-27.587510	63.290077	13.680117
34	-38.070308	-29.289980	61.180273	15.229492
35	-45.870932	-39.080010	70.493708	16.145710
36	-26.441001	-19.719978	66.724521	16.265422
37	-17.623449	-29.667885	91.342903	21.192094
38	-40.723148	-38.829945	60.317496	13.549624
39	-27.576841	-32.702908	59.096329	12.757344
40	-24.358193	-25.926277	75.674772	16.606097
41	-34.125178	-28.193876	64.412574	14.279508
42	-21.984864	-20.679716	54.021973	16.223316
43	-28.018306	-27.658524	68.196867	16.556379
44	-32.206515	-25.042350	60.008242	13.240342
45	-34.221808	-27.836763	80.371110	16.346321
46	-41.932709	-36.383377	62.216495	15.441866
47	-37.485330	-23.305546	60.712258	15.985380
48	-31.793870	-21.318190	53.310013	14.678392
49	-34.951017	-35.237827	65.509953	7.920439